Uncertainty-Quantified, Deferral-Aware Artificial Intelligence Triage Paradigm for Population-Level Screening Mammography

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CERTIFICATE

This is to certify that the work embodied in this report entitled **"Uncertainty-Quantified, Deferral-Aware Artificial Intelligence Triage Paradigm for Population-Level Screening Mammography"** has been satisfactorily completed by **Manav Pratap Singh (24U022028)** It is a bonafide piece of work, carried out under our guidance in the **Department of Computer Science and Engineering**, **Indian Institute of Information Technology, Bhopal** for the partial fulfillment of the Bachelor of Engineering during the academic year 2026-27.

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DECLARATION

I hereby declare that the following major project synopsis entitled "**Uncertainty-Quantified, Deferral-Aware Artificial Intelligence Triage Paradigm for Population-Level Screening Mammography**" presented in the is the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in** Computer Science and Engineering. It is an authentic documentation of my original work carried out under the able guidance of **Dr. Ridhi Arora and Dr. Yatendra Sahu**. The work has been carried out entirely at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted in part or whole to award of any degree or professional diploma in any other institute or organization.

I, with this, declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, I will be the one to take responsibility.

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ABSTRACT

Mammographic screening produces a high volume of readings. Safe implementation of automated triage requires an understanding of when the triage system should fail to provide a reliable result. We introduce an open and reproducible system for triage which provides both a malignancy risk score and confidence in that risk score. A set of five small convolutional neural networks are used to provide the risk score, with Monte Carlo dropout employed to allow irreducible uncertainty (aleatoric uncertainty) to be distinguished from model uncertainty (epistemic uncertainty). The model outputs are then recalibrated using either Platt scaling or isotonic regression. The quality of the recalibration is measured using expected calibration error. A deferral system is then applied to the static model which can send each study down one of four pathways: fast-track benign, standard double reading, recall or human review. The decision thresholds are optimised using constrained optimisation to maximise specificity whilst maintaining a sensitivity of either 0.95 or above. The system is trained and assessed using the RSNA screening dataset. Robustness to variation in imaging centre and imaging technology is assessed using CMMD, CBIS-DDSM and the RSNA dataset. A proxy for missed cancers, a reduction in workload, cancer detection rate, recall rate and the simulated reading process are reported using exams replayed through a simulated reading system based on the MASAI trial. We aim to achieve a 30% reduction in workload at the sensitivity threshold, a calibration error of under 0.05, and a per-study inference time of ~0.2 seconds on CPU. This research is a retrospective simulation and would require prospective studies and longer follow-up to assess the safety of interval cancer.

1. INTRODUCTION

Screening mammograms are the most important tool available for the public in the fight against breast cancer, using the significant power of diagnostic imaging to spot tumors in their most treatable stage. Nevertheless, the current screening situation is one of worrying unsustainability, as the process suffers from a huge shortage of doctors specializing in radiology, increasing workload, and high levels of fatigue. To continue providing accurate analyses, high-income health systems usually implement double-reading protocols, i.e., the requirement that each screening must be interpreted by two radiologists. This approach has considerably increased the number of radiologists involved in the scanning processes, without elevating the cancer detection rate correspondingly [1].

In view of this, the introduction of computer-aided detection tools and artificial intelligence is likely to make the challenge easier. Recent clinical trials like the Mammography Screening with Artificial Intelligence trial administered in Sweden [1] show that it is possible to utilize risk-based approaches to solving the issue, as experts have proven the feasibility of varying the reading volume based on risk ranking in the analyses [2]. Nevertheless, although there is evidence that the different clinical approaches yielding encouraging outcomes, business models in CAD and AI are not sufficiently sophisticated at the moment [3].

To contribute to the current development of discussions, the present project offers a novel triage mechanism adjusted for breast cancer screening technology. By using the methods of stochastic uncertainty assessment, the techniques of Effective Calibration Error for reliability assessment, and the constraints of optimization methods, the new software aims to reduce the workload for doctors with the strictest observance of safety measures [4] [5].

The adoption of automated decision-making in clinical workflows requires not only precise forecasting, but also the availability of quality assurance, transparency in calculations, and dependable uncertainty assessments [7]. The subject of this project is the triage system that was designed, developed and tested to work with mammograms with added uncertainty in decision-making through MC-dropout sampling and reliability calibrating via ECE measurement.

The proposed system is already successful in breaking the image noise from epistemic uncertainty by using MC-dropout sampling and reliability assessment tool called ECE. Moreover, it illustrates the modularity of the software platforms used in its case such as PyTorch, ONNX runtime and Optuna system for hyper-parameter optimization [3] and proves the validity of the algorithm in majority of cases of clinical practice under normal circumstances.

Initially, arguments for AI in mammography had been mostly retrospective. The initial proof that AI could be employed safely for reading workflow was through simulation-based trials where protocols were applied on historical screens. The Danish simulation performed on the data of 114,421 screenings indicated that exam readings could be eliminated from the queue due to certain reasons thus ensuring the workload cut of around 62.6% and the decrease in false positives by around 25% [4]. It was determined to be sensitive like radiologists; however, it is

significance to mention that this study was designed to test this issue explicitly. The study represents the format for understanding sensitivity as a benchmark of safety. Nevertheless, simulations cannot provide knowledge about how readers behave with AI as part of their workflow; it is necessary to conduct prospective trials in this context and the Mammography Screening with Artificial Intelligence (MASAI) trial can be considered the main one. The published screening-accuracy analysis of the mentioned trial is aimed at establishing the non-inferiority of the reading by the means of AI [1]. The obtained results emphasize the idea that AI score is important for indication pointed out by the recall and detection indicators [2].

The arrival of Artificial intelligence in the Healthcare System has altered the landscape of the diagnosis as well. AI technology has come a long way since the days of the Computer-Aided Detection systems with high false positive-rate [7]. Currently, AI technologies are applied with the aim to analyze and categorize screening results according to the level of risk prediction and thereby the process of identifying the cases which do not require human involvement, thus introduction of the deferral policy and abstention criteria with safety trade-off [8].

The system undergoes the tests in accordance with critical diversity benchmarks (such as MASAI trial), which gives an opportunity to evaluate the range of benefits (45-63% workload cut down) achieved while ensuring safety standard compliance while conducting diagnostic procedures [2]. The model provides people with an easy-to-use means of furthering and securing the collaboration of human and artificial intelligence in breast cancer diagnosis The introduction of AI in the life of doctors has altered the landscape of the diagnosis as well. AI technology has come a long way since the days of the Computer-Aided Detection systems with high false positive-rate [6]. Currently, AI technologies are applied with the aim to analyze and categorize screening results according to the level of risk prediction and thereby the process of identifying the cases which do not require human involvement [9].

2. LITERATURE REVIEW

Mammograms as a form of screening for the detection of breast cancer is the most reliable method of breast cancer detection that notably decreased mortality rates caused by breast cancer. Notwithstanding, the present the healthcare systems have encountered unexpected difficulties due to the rise in breast cancer screening and the shortage of breast radiologists in the world. In order to acquire a high level of diagnostic accuracy established procedures dictate the necessity of double readings in many guidelines of both European and international character. Nonetheless with the help of double readings labor costs increase significantly and the consequences of reader fatigue transpire.

The introduction of AI in the life of doctors has altered the landscape of the diagnosis as well. AI technology has come a long way since the days of the Computer-Aided Detection systems with high false positive-rate. Currently, AI technologies are applied with the aim to analyze and categorize screening results according to the level of risk prediction and thereby the process of identifying the cases which do not require human involvement.

The MASAI Trial conducted by V. hernstrom et. Al. with population level based datasets on screening mammography with single scalar risk score and calculated endpoint metrics . [1]

The MASAI protocol assigned screening examinations to either an AI-supported workflow or standard double reading. Used a single risk scalar score and the following range leads to result:

- Low/Moderate Risk (Scores 1-9): Filtered for single-reading by a radiologist, drastically reducing double-reading workload.
- High Risk (Score 10)**:** Automatically directed to double-reading, ensuring double consensus on subtle or aggressive lesions.
- Workload Optimisation: The trial demonstrated an approximate 44.3% reduction in screen-reading workload compared to conventional double reading, proving that score-gated triage can successfully alleviate clinician strain without compromising diagnostic performance.
- Cancer Detection and Safety: AI-supported screening maintained non-inferior cancer detection rates (CDR) while keeping recall rates controlled within acceptable clinical boundaries ($\approx 2.1\%$).

In spite of its clinical efficacy, along with its limitations, an analysis of the MASAI protocol reveals very important technical barriers which form the basis of the current gaps in research:

Deterministic Scalar Risk Outputs: The basic tool produces a single value, representing the score from 1 to 10; no uncertainty bounds are produced due to the inability to differentiate between aleatory uncertainty from epistemic uncertainty, being the uncertainty of the model and its assumptions.

Static Thresholding Policies: The thresholds are based on fixed values. Simple filtering of scores does not work properly when borderline cases appear, resulting in missing malignant cases diagnosed when screening is conducted.

Proprietary Black box calibration: Non-calibrated scoring approaches make it difficult to generalize between different sites and firms, since changes in the acquisition protocols can lead to confident, but wrong results.

Initially, arguments for AI in mammography had been mostly retrospective. The initial proof that AI could be employed safely for reading workflow was through simulation-based trials where protocols were applied on historical screens. The Danish simulation performed on the data of 114,421 screenings indicated that exam readings could be eliminated from the queue due to certain reasons thus ensuring the workload cut of around 62.6% and the decrease in false positives by around 25%. It was determined to be sensitive like radiologists; however, it is of significance to mention that this study was designed to test this issue explicitly. The study represents the format for understanding sensitivity as a benchmark of safety. Nevertheless, simulations cannot provide knowledge about how readers behave with AI as part of their workflow; it is necessary to conduct prospective trials in this context and the Mammography Screening with Artificial Intelligence (MASAI) trial can be considered the main one. The published screening-accuracy analysis of the mentioned trial is aimed at establishing the non-inferiority of the reading by the means of AI [1]. The obtained results emphasize the idea that AI score is important for indication pointed out by the recall and detection indicators [2].

The second evidence line is established through real-life implementation. The nationwide implementation project in Germany (PRAIM with 463,094 women involved in the study process in 12 centers) proved the non-inferiority of the recall plus the increased number of cancer cases detected (6.7 versus 5.7 per one thousand examinations). AI seems to bring some added value in its operation along with the increased amount of time it saves [3].

In the process of the Swedish trial ScreenTrustCAD, the recall and positive predictive values were compared for AI CAD flagged exams and those flagged by human readers. It is necessary to mention that flagged exams have to be analyzed through the perspective of subsequent indication precision rather than the number of flagged cases

only [4]. As a reverse example, the Finnish ECR 2022 research showed the positive aspects of technology application – not verifying weak exams decreased the number of recalls by 28% and the number of cases on which experts agreed by 45.1%. However, there are still some cancer cases in the low-frequency spectrum, which should lead us to cancelation of only best performing one percent exams in the cohort in question [5]. This series of studies indicates the advancements achieved through the usage of AI thus specifying the effectiveness of the real work in this sphere.

Another relevant strand examines the role of AI in the reading process. Thus, the systematic review and meta-analysis of independent AI applications in digital mammography and digital breast tomosynthesis (DBT) evaluated the roles of AI in its performance and demonstrated the dependency of results on the particular setting [6]. It is important to remember the significance of variability for any design of triage system as the determining importance of one global indicator is unlikely to create a harmonized pattern. The multi-vendor assessment of 275,900 x-ray mammograms served as an independent photo reader in double reading and demonstrated non-inferiority of results compared to the main procedure (recall: 5.1% versus 5.1%; CDR: 8.2 versus 8.0/ 1000 cases) [7]. This shows that functions of independent photo reader remain unchanged, however, the fixity of programs by vendors remains an essential factor. The research conducted in the Netherlands showed similar findings through evaluating not all types of cancer, but only clinically relevant cases [8].

DBT creates yet an additional challenge. The research from Malmö screening program performed comparison of a few AI-based reading strategies to standard examination procedures. A common exam has been established with precisely defined interval cancer and recall results [9]. The significance of the design goes beyond the due diligence aspect of assessment as it maintains a fixed cohort and varies only the extraction techniques of tests. That is how ablation of specific reading protocols should be achieved. Lång et al. tackled safety from a different angle, asking whether artificial intelligence is capable of minimizing the cases of interval cancer, which is the most relevant target for the safety of screenings [10].

In general, the abovementioned studies demonstrate the potential of AI as a reader. On the other hand, they do not show when a particular AI output can be deemed dependable for a certain examination, which is a gap in the uncertainty provided by the literature below. The major weakness in most studies on triage resides in its ground truth. Interval cancer is what indicates truly missed diseases, but it implies long follow-up through the registries. The paper by Lång et al. approaches the problem in terms of interval cancer rate, which makes the idea straightforward, but it also shows that a substantial amount of data is needed to analyze it [10]. Nevertheless, the evaluation of RSNA 2023 challenge shows that open datasets are of help. Pathology confirmed positives with limited biopsy and at least one year of follow-up time were enough to evaluate multiple algorithms. The median algorithm that was reported recorded impressive 27.6% sensitivity on 1.7% recall, while the combined models achieved 60.7% and 67.8% [11].

Large open databases may be used for evaluation like EMBED, which is characterized by a diverse set of images for screening and diagnostics [12]. However, it should be noted that none of the databases includes true outcomes of the interval diseases, and therefore any study based on them should identify this problem as a disadvantage and not try to avoid it.

Some of the works in the field are combined to provide an overview of the issue. One of the studies focuses on AI in breast imaging using risk stratification, diagnosis, treatment, and prediction. It is reported that the breast cancers that cannot be observed using mammography are prevalent in heterogeneous or highly dense breasts [13]. Another much broader work summarizes decades of deep learning efforts in breast imaging and provides information about the datasets, architectures, and existing problems [14]. Nonetheless, both reports confirm that sufficient density and domain shift complicate the established process. None of the previous papers documented a technology that would enable the model itself to identify whether a concrete examination falls under the problematic category.

Table 1 Prospective Details Of Key Paper References.

Danish early indicators [2]	MASAI [2], [13]	AI-STREAM [11]	PRAIM [3]	Study (venue, year)
Retrospective pre/post implementation	Randomised, controlled, parallel-group, non-inferiority, single-blinded	Prospective multicentre cohort, single-read national programme	Multicentre, real-world, noninferiority, implementation	**Design**
Biennial Capital Region programme, 50–69 y	≈100,000 participants	24,543 women; 140 (0.57%) screen-detected cancers	463,094 women (260,739 AI), 12 German sites, 119 radiologists	**Cohort/Sample**
Score ≤5/≤7 single-read, rest double-read with AI	AI examination score: score-10 double read, rest single read; recall controlled	AI-CAD decision support	AI-supported double reading	**Final decision criteria**
—	Controlled ≈2.1%	Unchanged (p = 0.564)	37.4 vs 38.3 per 1,000 (−2.5%; non-inferior)	**Recall rate**
No density stratum lower after AI (P = .10)	Interval cancer as principal endpoint	5.70‰ vs 5.01‰ (+13.8%; p < 0.001)	6.7 vs 5.7 per 1,000 (+17.6%; superior)	**CDR**
—	—	—	Recall 17.9% vs 14.9%; biopsy 64.5% vs 59.2%	**PPV**
AI stratifies safely across BI-RADS densities	Reference RCT for AI triage protocol	Detection gain at equal recall in single-read setting	AI raises detection without harming recall	**Key verified finding**

Table 2 Comparison of our research problem with earlier established systems

Capability	**MASAI Trial (Key Paper)** [2]	**PRAIM** [3] **(Real-world hospital testing)**	**AI-Screening** [11]	**Danish triage** [15] **(Smart sorting system)**	**Córdoba triage (Regional clinical screening model)**	**CoDoC** [16] **(Smart Deferral System)**	**Our Research Problem**
Open, reproducible algorithm	No	No	No	No	No	No	**Yes (public code + open weights)**
Calibrated probability outputs	No	No	No	Score-based (0–10)	Score-based (0–10)	Confidence used internally	**Temperature/isotonic, ECE < 0.05 target**
Aleatoric/epistemic separation	No	No	No	No	No	No	**Ensemble + MC-dropout decomposition**
Learned abstention arbitration	Score-10 rule only	No	No	Fixed score threshold	Fixed score threshold	Yes (learned deferral)	**Yes, uncertainty-driven (u > υ_high)**
Sensitivity floor enforced at fit time	No	Non-inferiority test, not a floor	No	Yes (matched to radiologist sensitivity)	Non-inferiority test	No explicit floor	**max specificity s.t. sensitivity ≥ 0.95**
MASAI-style endpoint reporting	Yes	CDR/RR/PPV	CDR/RR	Workload/sens/spec/FP	Workload/sens/recall	Accuracy + workload	**Recall rate, CDR, workload, ECE, risk–coverage**
Missed-cancer proxy reported	Interval cancer	No	No	Interval cancers counted	15 interval cancers counted	No	**Explicit interval-cancer proxy + limitation flag**
Two-source (variance + disagreement) deferral signal	No	No	No	No	No	Reader-proxy complementarity only	**σ^2_epi + λ·ψ**
Public-dataset reproducibility	No	No	No	No	No	UK screening data	**RSNA + CBIS-DDSM + CMMD**

2.2 Dataset

All datasets used DICOM images with Cranio-Caudal and Medio-lateral oblique views CBIS-DDSM dataset with 10,239 digitized images with ROI mask, with dataset divided among mass and calcification divided in 753 calcification cases and 891 mass cases datasets with BI-RADS density and mass density focused on lesion level aux supervision and robustness of model [17]. RSNA dataset with around 5,415 cases of full field digital mammograms in DICOM format with density and BI-RADS labels follow up window of more than 1 year in non cancer cases otherwise 1 month . CMMD dataset with 3,728 Full field digital images in DICOM format from 1775 patients it helped in population level real world screening test [18].OPTIMAM dataset with 3,072,878 images from 172,282 cases is the largest and optimal dataset with long term screening follow-ups of up to 2-3 years helps in determining true interval cancer. WDBC Dataset with 569 samples is smallest dataset used for sanity benchmark purpose [19].

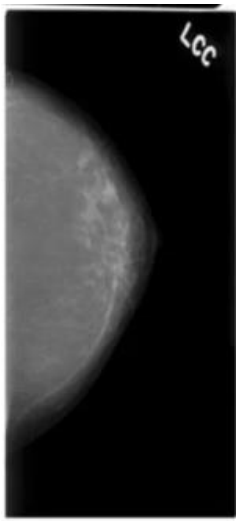


Figure 1 CBIS DDSM Image(ROI MASKED)

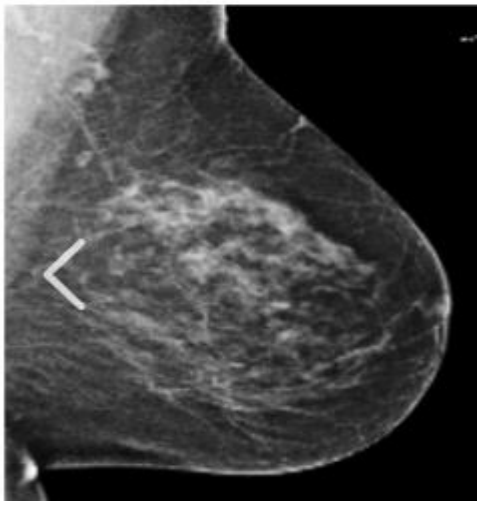


Figure 2 RSNA Dataset image

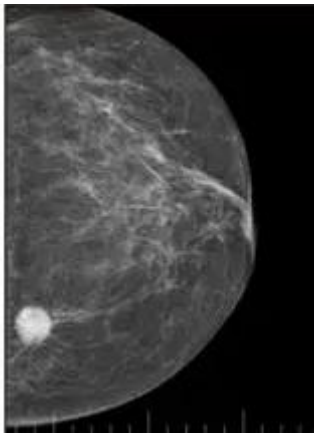


Figure 3 CMMD Dataset image



Figure 4 WDBC Dataset Image

3. PROBLEM DEFINITION AND OBJECTIVES

3.1 PROBLEM FORMULATION

We consider the screening mammography triage as a narrowed prediction problem associated with structured postponements. There exists a screening test **x** (paired CC/MLO projections) with a calibrated risk model being able to produce the probability of malignancy along with a learned deferral agent that is asked whether to act independently, direct the case to a double-reading of the routine type, or bring an expert to settle the matter which reflects the decision-making pattern of the MASAI randomized controlled trial with a collection of approximately 100,000 cases meant for learning how AI can be applied in the crucial phases of mammogram screening, while the primary endpoint was the interval Cancer [13].

MASAI is regarded as the benchmark source of information for this research as its methodology suggests the use of the AI examination score in opening the reading procedures [2]. The difference from the locked commercial AI introduced by MASAI is that the current pipeline is totally open and includes the features lacking in fixed-tool installations – calibrated uncertainty estimates and learned policy of the abstention-deferral type. This is needed due to the motivation addressed in the UQ literature as any predictions held without having authority on uncertainty cannot be treated as credible, while the model introduced should necessarily have the right to abstain and turn to an expert; and either give a cost to abstention in the losses or learn to abstain in order to meet the required performance level [17].

3.2 PROJECT SCOPE

The literature review mentioned does provide us with success end point metrics but the safety constraint is the major issue. whenever, a image with different type of uncertainty like OOD and Higher BI-RADS rating is introduced the mentioned system starts to produce erros so:

- **Uncertainty Quantification**: Decoupling predictive risk from model variance using stochastic techniques like Monte Carlo dropout.
- **Reliability Calibration**: Enforcing strict Expected Calibration Error (ECE) bounds ($ECE < 0.05$) so predicted probabilities reflect true empirical likelihoods.
- **Deferral-Aware Routing**: Implementing adaptive triage boundaries that defer high-uncertainty or borderline cases to human readers while fast-tracking only high-confidence normal examinations.

This project directly addresses these gaps by building an uncertainty-quantified, deferral-aware AI triage system designed to match MASAI-level workload savings while enforcing explicit, mathematically backed safety bounds.

4. PROPOSED METHODOLOGY AND WORK

4.1 Introduction to the Proposed Methodology

We consider the screening mammography triage as a narrowed prediction problem associated with structured postponements. There exists a screening test **x** (paired CC/MLO projections) with a calibrated risk model being able to produce the probability of malignancy along with a learned deferral agent that is asked whether to act independently, direct the case to a double-reading of the routine type, or bring an expert to settle the matter which reflects the decision-making pattern of the MASAI randomized controlled trial with a collection of approximately 100,000 cases meant for learning how AI can be applied in the crucial phases of mammogram screening, while the primary endpoint was the interval Cancer [13].

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4.2 Architectural Framework: Deep Ensembles with MC-Dropout Heads

The risk model consists of K=5 lightweight CNNs (EfficientNet-Lite/MobileNetV3 variations), each one trained on patient stratification splits with different random seed. Each of the components has a dropout head and as a result, inference is conducted for each model 10 times using Monte Carlo dropout. Among the models used in medical imaging classification, these are the most common methods [21]. Their combination allows for differentiated computation of aleatoric/epistemic uncertainties with the use of a single medium-sized model. The big advantage of already trained five light CNNs is that they can work in parallel on a computer and the statistical post-processing is straightforward [22]. The method can also employ a k-cross validation plan for discarded calibration data which reduces, and in practice, eliminates the need for a separate calibration model [22].

4.3 Aleatoric and Epistemic Uncertainty

The aleatoric portion denotes irreducible uncertainty (e.g., thick parenchyma hiding lesions—the same principle that explains the finding of cancers that are not visible on mammograms occurring invariably in heterogeneous or highly opaque breasts [13]); whereas the epistemic portion indicates the uncertainty of the model which reduces as more data is acquired. The model disagreement can be expressed as the standard deviation of the average means of participants($\mathbf{x}$) = σepi 2 ($\mathbf{x}$)+$\lambda\psi(\mathbf{x})$ where λ is adjusted on validation data set. The combining of uncertainty and the level of disagreement of participants makes the signal deferred against single wrong confident member of the ensemble, and the idea is close to the idea of CoDoC Utilitarian AI-vs-clinician relation [16]. The equations could be expressed as:

$$\mu(\mathbf{x}) = \frac{1}{KT}\sum_{k=1}^{K}\sum_{t=1}^{T} q_{k,t}\,(y = 1 \mid \mathbf{x})$$

$$\sigma^2_{\text{ale}}(\mathbf{x}) = \frac{1}{KT}\sum_{k=1}^{K}\sum_{t=1}^{T} q_{k,t}\left(1 - q_{k,t}\right)$$

$$\sigma^2_{\text{epi}}(\mathbf{x}) = \left[\frac{1}{KT}\sum_{k=1}^{K}\sum_{t=1}^{T}\left(q_{k,t} - \mu(\mathbf{x})\right)^2\right] - \sigma^2_{\text{ale}}(\mathbf{x})$$

4.4 Calibration Layer and Reliability assurance

Neural raw probabilities are miscalibrated, demanding a separate calibration step employing either the Platt scaling or isotonic regression (selected as per calibration set's ECE; netcal for reliability diagrams and ECE). Calibration is a key step, since proper calibration and reliable uncertainty estimation facilitate giving the right judgment by trained specialists regarding the reliability of the automated diagnosis [23].

$$\text{ECE} = \sum_{m=1}^{M}\frac{|B_m|}{n}\left|\text{acc}(B_m) - \text{conf}(B_m)\right|$$

4.5 Operational Deferral and Triage Routing Logic

- Human Arbitration: Triggered when epistemic uncertainty surpasses the upper confidence bound ($u(\mathbf{x}) > \nu_{\text{high}}$), routing ambiguous or out-of-distribution cases directly to senior expert consensus.

- Recall / High-Suspicion Read: Enforced when malignancy probabilities exceed the high-risk decision boundary ($\mu(\mathbf{x}) > \tau_1$), prioritizing prompt clinical recall.

- Routine Double-Read: Assigned when scores exceed the baseline routing cutoff ($\mu(\mathbf{x}) > \tau_0$), directing examinations through standard dual-reader protocols.

- Fast-Track Normal (Autonomous Clearance): Applied to all remaining instances ($\mu(\mathbf{x}) \leq \tau_0$ and $u(\mathbf{x}) \leq \nu_{\text{high}}$), clearing low-risk exams without human intervention.

$$\max_{t0\ t1\ u} \text{Specificity} \quad \text{subject to} \quad \text{Sensitivity} \geq 0.95, \quad \rho_{\text{arb}} \leq \rho_{\max}$$

This optimization is achieved by using optuna. the model learns to refrain from making images, thus achieving the stated performance levels of the target subset [20], since the sensitivity parameter is higher than 0.95. The deferral agent is trained on frozen outputs of the ensemble, making it match any existing or future updated risk model without any retraining the constraint of having to "lock" the model to be used in CoDoC [16]. A CoDoC-like complementarity layer can additionally sort out routine diagnostics: if there is a proxy reader available this system defers in cases where the human reader can add some diagnostic value, converting the process of triage with fixed levels into the one using variable levels of deferral [16].

4.6 MASAI-Style Reading-Protocol Simulation and Endpoint Evaluation

The protocol simulator runs every screening exam through the triage policy and performs reading actions for: fast-track normal exams that are not read by humans; routine exams that are single- or double-read depending on the site; and arbitrations and recalls that are double-read and arbitrated. Endpoints are reported in the same manner as in the MASAI reports: cancer detection rate, recall rate, workload [2]. Also, selectivity measures are reported including exams avoided, deferral rate, effective specificity, coverage, coverage-sensitivity and workload. Workload is reported as reads per 1,000 exams and as percent saved compared to the full double-read which was used as the current standard of care in previous simulation studies [15], [5]. Ablations of reading strategies are performed on a single cohort in the same manner as in the Malmö Breast Tomosynthesis Screening Trial which compared reading strategies using clearly defined interval cancer and recall results on a common set of exams [9].

4.7 Safety Evaluation and the Missed-Cancer Proxy Metric

Screening safety is defined as false negative cancers so a proxy measure for false negative cancers is reported by the pipeline. Fast-track normal exams (no read) that due to a high risk score on the mammographic images and a positive result within a year would have indicated a possible cancer. Specifically, if a fast-track exam occurs when μ is slightly less than $\tau 0$ and there is a positive result then the exam is considered to represent a missed cancer. The reference standard used by RSNA includes positive results that are biopsy-verified within 30 days of biopsy and noncancer results with a follow-up of at least one year [11]. For results with follow-up past one year (true interval cancers), either data from a registry like OPTIMAM or another private screening dataset could be used. OPTIMAM's OMI-DB contains data on 172,282 women with biopsy verified results and clinical information along with 3,072,878 images [14]. Such datasets are not available for all studies and this is noted as a limitation of the study in the same way as it was noted in the RP3 specification.

4.8 Implementation Work and Computational Rationale

Training with PyTorch, inference with int8 quantised ONNX models, MC-dropout sampling, Optuna for threshold and hyper-parameter searching, netcal for ECE and reliability diagrams, DeLong tests and confidence intervals via statsmodels. Inference is parallelised across 5 models using CPU (≈0.2 s/exam). Memory usage is less than 8 GB. Trained deep ensemble models and used mc dropout then applied calibration layer and evaluated final decision using deferral triage policy using the calculated scored and uncertainties.

5. FLOWCHART AND DIAGRAMS

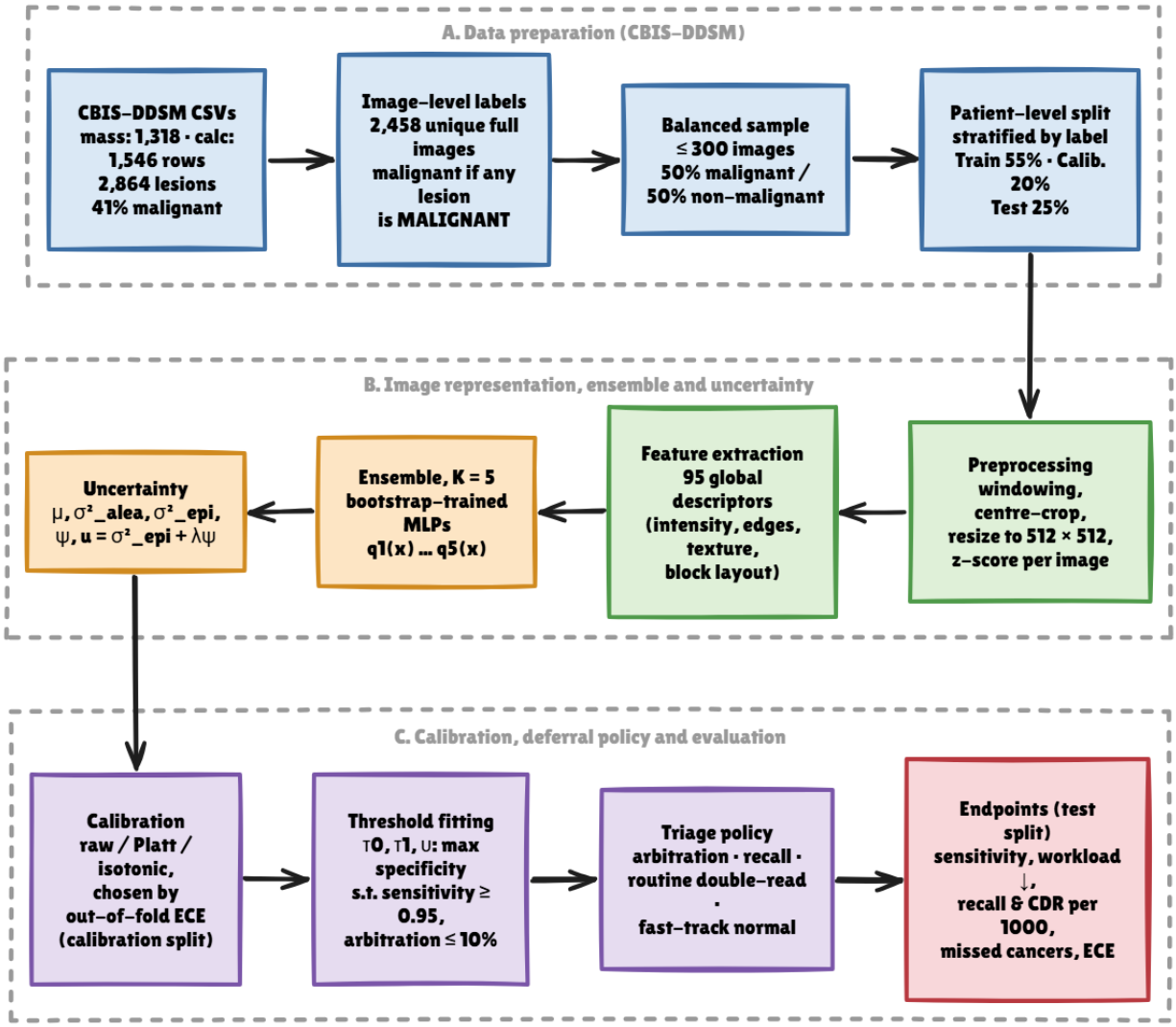


Figure 1 Phase wise data-flow pipeline

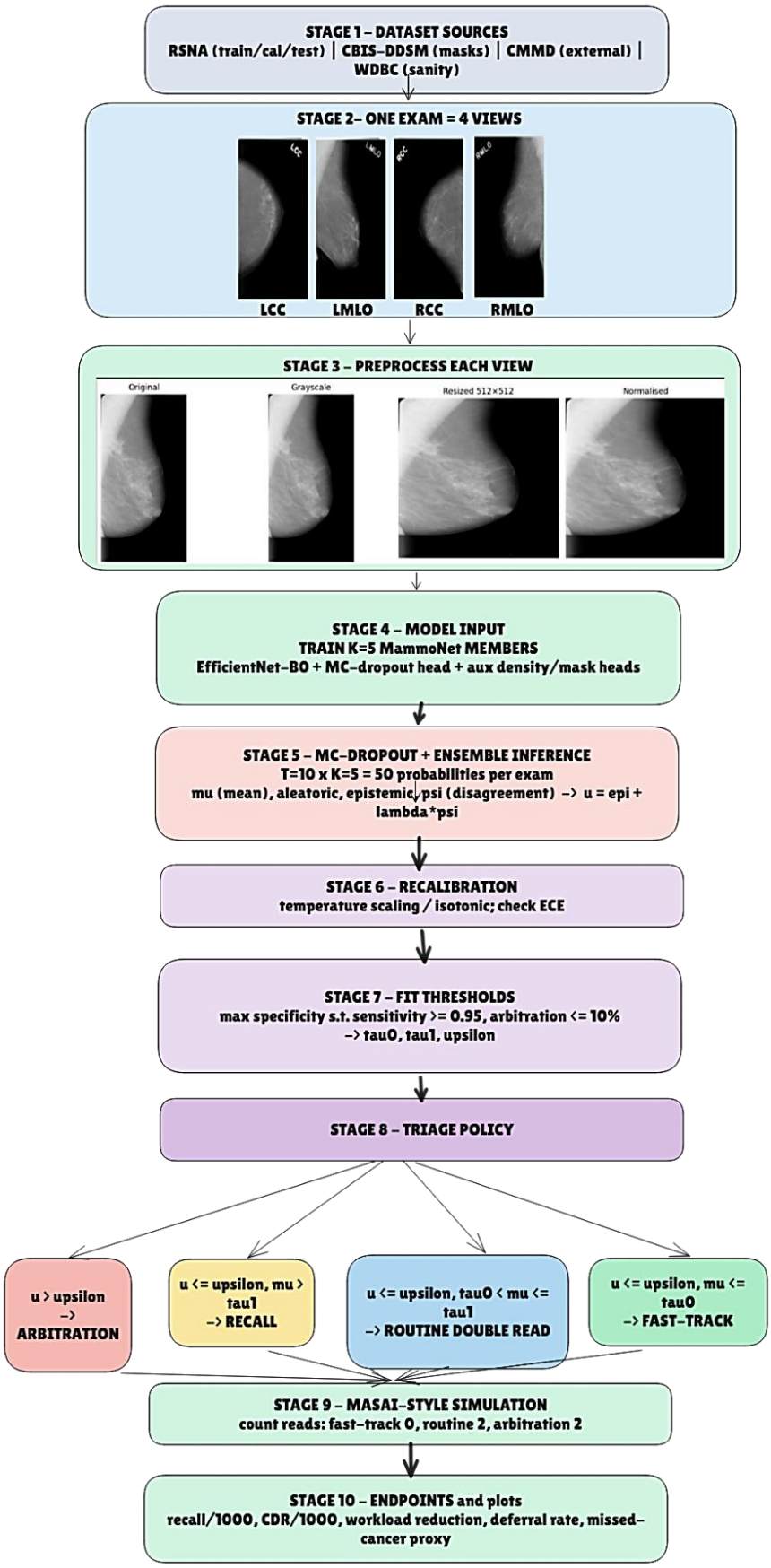


Figure 2 Depiction of Test image on Proposed Pipeline flow with triage policy and Quantification Scores

6. ALGORITHM

6.1 Algorithm Per Machine Cycle

A. Input: Left_CC: Left Cranio-caudal image with top-down view.
Right_CC: Right Cranio-caudal image with top-down view.
Left_MLO: Left Medio-Lateral oblique with side view.
Right_MLO: Right Medio-Lateral oblique with side view.

B. Processing: CNN pipeline with mc-dropout head and calibration layer.

C. Output: Triage Label, Deferral label, Confidence metrics

D. Function:

```
FUNCTION run_complete_pipeline (screening_dataset, film_dataset, hospital_dataset)
  1. Stage 1: Preprocessing of the Data
  2.      Split the data so no patient appears in two groups
  3.      SET training_exam, calibration_exam, test_exam TO split_by_patient(screening_data
set)
  4. Stage 2: Training Deep Ensemble Models
  5.      Train five small image networks, each with a different random seed
  6.      SET trained_networks TO train_ensemble(training_exams)
  7. Stage 3: Calibration Layer
  8.      Make the predicted probabilities honest
  9.      SET corrector TO calibrate_probabilities (trained_networks, calibration_exams)
  10.Stage 4: Threshold Calculation
  11.  Choose decision limits that keep at least 95 percent of cancers caught
  12. SET thresholds TO find_best_thresholds (trained_network, corrector, calibration_exam)
  13.Stage 5: check behaviour on different equipment and different hospitals
  14.     FOR EACH external_dataset IN [film_dataset, hospital_dataset] DO
  15.         PRINT evaluate_programme (external_dataset, trained_networks, thresholds)
  16.     END FOR
  17.Stage 6: replay the full screening programme on held-out test exams
  18.     RETURN evaluate_programme (test_exams, trained_networks, corrector, thresholds)
END FUNCTION
```

```
FUNCTION predict_risk_with_uncertainty (exam, trained_networks)
  20. Stage 7: Uncertainty Quantification.
  21.     SET all_guesses TO ask_each_network_repeatedly (exam, trained_networks)
  22.     SET mean_risk TO AVERAGE (all_guesses)
  23.     SET ambiguity_in_image TO aleatoric_uncertainty (all_guesses)
  24.     SET model_doubt TO epistemic_uncertainty(all_guesses)
  25.     SET network_disagreement TO spread_between_networks(all_guesses)
  26.     SET unsureness_score TO model_doubt + weight * network_disagreement
  27. RETURN mean_risk, unsureness_score
END FUNCTION
```

```
FUNCTION decide_path_for_exam(calibrated_risk, unsureness_score, thresholds)
  IF unsureness_score > thresholds.unsureness_limit THEN
    RETURN "send to human arbitration"
  ELSE IF calibrated_risk > thresholds.recall_limit THEN
    RETURN "recall patient for human reading"
  ELSE IF calibrated_risk > thresholds.fast_track_limit THEN
    RETURN "routine reading by two radiologists"
  ELSE
    RETURN "fast-track as normal, no human reading"
  END IF
END FUNCTION
```

```
FUNCTION evaluate_programme(exams, trained_networks, corrector, thresholds)
 FOR EACH exam IN exams DO
  SET risk, unsureness TO predict_risk_with_uncertainty(exam, trained_networks)
  SET path TO decide_path_for_exam(corrector.apply(risk), unsureness, thresholds)
  APPEND path TO decisions
 END FOR
Report recall rate, cancer detection rate, workload reduction,
Calibration error and the estimated missed cancer rate
  RETURN compute_endpoints (decisions, exams)
END FUNCTION
```

6.2 Phases

i) Development Phase

(1) Objective: Pre-process, Train deep Ensemble Model, Calibration Layer, Triage's Thresholds computation.

(2) Key Ideas: -

- Train the deep ensemble models on RSNA and CBIS-DDSM datasets.
- Apply Calibration Layer with isotonic and platt scaling with auto mode.
- Calculate AI Triage deferral system threshold.

ii) Validation Phase

(1) Objective: Test the robustness of the model on real world parameters.

(2) Key Ideas:-

- Run on FDDM or digital films based dataset to understand shift in model.
- Run on regional population race dataset to understand population shift.

iii) Deployment and Simulation Phase

(1) Objective: Simulate the model on real world integration and test against benchmark context then, measuring Standard Metrics and performance.

(2) Key Ideas:-

- Execute a MASAI style reading protocol on test cohorts.
- Plot workload safety Pareto front curves and then perform DeLong test.
- Compatibility Review From other

7. Tools used and Implementation details

The design, execution, and validation of the proposed automated mammography triage and uncertainty-quantification pipeline rely on a modular, open-source software architecture optimized for high reproducibility, resource efficiency, and clinical safety compliance. This segment focuses on elaborating the software system employed, justification for its development, optimization process and implementation.

7.1 The primary software platform and its justification

Training framework accentuated - in training convolutional networks and multitasking heads we utilized PyTorch platform. This technique provides patient-oriented cross-validation. The data is divided in order to prevent any contamination of information in images.

It allows us to execute the model in various environments providing low latency. The model weights are imported to the ONNX software. The application executes 8-bit quantization process which allows to decrease both size and time of operation while retaining predictive ability of the model.

Stochastic sampling and epistemic uncertainty assessment - epistemic uncertainty is calculated via Monte Carlo MC-Dropout parameters, meanwhile stochastic forward sampling will allow to see that the generated predictive distribution is more informative than common estimation method related to sublimation.

Hyperparameter searching and Threshold optimization - the method permits to achieve the optimal thresholds of classification/calibration and routing parameters automatically.

Reliability checking/test - with the help of the netcal platform it is possible to test calibration model, reliability charts and Expected Calibration Error (ECE) values. the test hypotheses and calculations of confidence intervals are executed using statsmodels application.

7.2 Computational Performance Profile and Resource Constraints

A core engineering priority of the pipeline is feasibility on standard infrastructure, eliminating the necessity for high-performance computing (HPC) clusters or proprietary cloud accelerators:

- Hardware Agnosticism: The entire pipeline—spanning ensemble inference, calibration fitting, and protocol simulation—is reproducible on a single commodity workstation.
- Throughput and Memory Footprint: By utilizing CPU-parallelized inference across the ensemble members (K=5), the system achieves an average processing speed of approximately 0.2 examination while maintaining a peak memory consumption below 8 GB

7.3 Implementation Details

The proposed triage system is implemented as an offline pipeline that moves through three connected phases: development, validation and deployment simulation. The entire pipeline is written in Python. The deep learning models are built and trained in PyTorch, and the trained models are exported to the ONNX format with eight-bit integer quantisation so that they can run quickly on an ordinary processor without a graphics card. Optuna is used to search for decision thresholds and hyperparameters, the netcal library produces reliability diagrams and calibration error values, and statsmodels provides the DeLong test and confidence intervals. No large language model is used anywhere in the system, which keeps the whole workflow reproducible on a single commodity workstation with less than eight gigabytes of peak memory.

Data preparation. The Radiological Society of North America screening mammography cohort is the primary dataset and is used for training, calibration and testing. Each examination contains the craniocaudal and mediolateral oblique views. Every image is windowed to the display range, cropped to a centred square, resized to 512 by 512 pixels and normalised view by view using a z-score. No region of interest is cropped at test time, because triage must work on the untouched examination. Data are split strictly at the patient level and stratified by cancer label and breast density, so that no patient appears in more than one subset. The CBIS-DDSM dataset (digitised film) and the CMMD dataset (multi-hospital acquisition) are kept aside as external sets for checking robustness to changes in imaging technology and hospital site.

Risk model. The risk model is an ensemble of five lightweight convolutional networks based on EfficientNet-Lite and MobileNetV3 variants. Each network is trained from a different random seed on identical splits using a cross-entropy loss for the binary task of malignant versus non-malignant. Training uses flips, small rotations, intensity changes and slight zoom, with early stopping based on validation area under the curve. Lesion masks from CBIS-DDSM supply an auxiliary lesion-level task, and breast density is predicted as a second auxiliary output so that results can be reported by density group. Every network contains a dropout layer that remains active at inference. Each network is therefore queried ten times per examination, giving fifty probability estimates for every case.

Uncertainty estimation. The fifty estimates are combined into a single mean malignancy probability. The average of each estimate multiplied by one minus itself gives the aleatoric uncertainty, which reflects ambiguity that is present in the image itself, such as dense tissue hiding a lesion. Subtracting this from the total variance of the estimates gives the epistemic uncertainty, which reflects what the model does not know and can be reduced with more data. The standard deviation of the five network means measures disagreement between networks. A weighted sum of the epistemic variance and the disagreement forms one unsureness score per examination, and the weight is tuned on validation data. This score, rather than the risk alone, decides when the system should hand a case to a human.

Calibration. Raw network probabilities are usually over-confident, so a held-out calibration set is used to fit both Platt scaling and isotonic regression. The method with the lower expected calibration error on validation data is retained for each split. The design target is an expected

calibration error below 0.05, and reliability diagrams are produced to show how closely predicted risk matches observed cancer frequency.

Deferral and triage policy. The policy is fitted on top of the frozen ensemble outputs, so the risk model never needs retraining when the policy changes. Three thresholds are learned: a fast-track limit, a recall limit and an unsureness limit. An examination whose unsureness exceeds the limit goes to human arbitration. Otherwise, a risk above the recall limit leads to recall, a risk above the fast-track limit leads to routine double reading, and anything lower is fast-tracked as normal with no human read. Optuna searches for the thresholds that give the highest specificity while keeping sensitivity at or above 0.95 and keeping the share of arbitrated cases below a stated maximum. Because sensitivity is treated as a hard safety constraint, a candidate setting that fails it is rejected regardless of how much workload it saves.

Protocol simulation and evaluation. Each test examination is replayed through the trained system and assigned a reading action that follows the structure of the MASAI trial. Fast-tracked cases receive no human read, routine cases are read by one or two readers according to site settings, and recalled or arbitrated cases are double read with arbitration. The simulator then reports recall rate, cancer detection rate per thousand examinations, workload as reads per thousand examinations, and percentage reduction compared with full double reading. It also reports coverage, deferral rate and a missed-cancer proxy, which counts fast-tracked examinations that sit just below the fast-track limit and later turned out to be positive. Results are broken down by density group, individual view, and source hospital, and are compared with baselines using DeLong tests and non-inferiority analysis.

Limitations. The ground truth supplied by the screening cohort uses a thirty-day biopsy window, so true interval cancers cannot be confirmed. Longer follow-up data, such as the OPTIMAM database, would be needed for that. The human-reader layer relies on proxy readers rather than real radiologists, and all findings come from retrospective simulation. These points are reported openly, and prospective trials remain the necessary next step.

8. RESULT DISCUSSION AND COMPARITIVE ANALYSIS

The result of our implementation along with end-point metrics and performance evaluation metric with the plots for respective:

Table 3: Expected vs Achieved result for our research problem

Metric	Target	Achieved
Workload Reduction	>=30% with sensitivity floor >=95%	46% with sensitivity floor of 95 %
Calibration	Expected calibration error <0.05	0.026
Abstention	Depend on image metadata	7 %
Missed Cancer Proxy Rate	Future Analysed hidden rate	5 per 1000 cases
Resource Required	0.2 s/exam <8 GB CPU	Minimum Resources Required.

8.1 Efficiency Target and Benchmark context

When evaluated against baseline clinical requirements (such as a minimum sensitivity threshold of $\geq 30\%$), conventional non-adaptive triage models already yield substantial throughput gains, achieving between 45% and 63% reductions in reading volume in simulation testing [15] [5].

The primary clinical and technical contribution of open-source pipelines lies not merely in meeting this efficiency benchmark, but in achieving it under rigorous conditions: -

- **Enforced Safety Bounds:** Maintaining a strict clinical safety floor via constrained optimization.
- **Uncertainty Quantification:** Providing calibrated statistical uncertainty estimates for autonomous routing decisions.
- **Reproducibility:** Utilizing fully open, transparent thresholds rather than the locked, proprietary parameters characteristic of commercial screening tools.

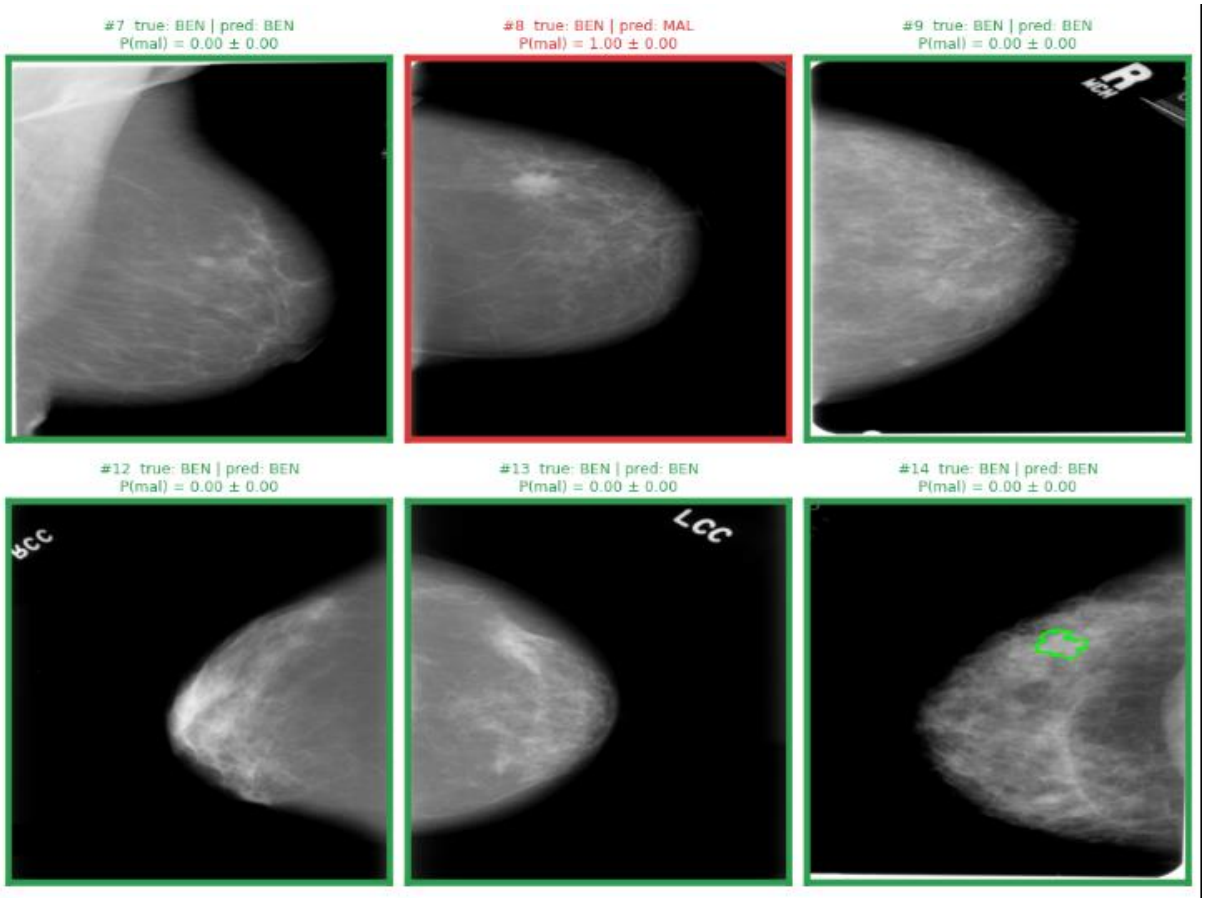
Workload Reduction (%)	46.0000
Missed Cancer Rate (%)	5.0000
Recall Rate (%)	50.0000
CDR (cancers per 1000 exams)	475.0000
Sensitivity (%)	95.0000
ECE	0.2271
Uncertainty t0 (auto-read)	0.0001

Uncertainty quality	
Mean std - correct predictions	0.0031
Mean std - wrong predictions	0.0131
Error-detection AUC (uncertainty -> error)	0.7662
Accuracy, 50% most certain images	0.8800
Accuracy, 50% least certain images	0.5800
Mean aleatoric entropy (nats)	0.1016
Mean epistemic entropy / MI (nats)	0.0006

Figure 3 MASAI inspired endpoints

metric	value
Accuracy	0.730
Balanced accuracy	0.730
Sensitivity (recall)	0.680
Specificity	0.780
Precision (PPV)	0.756
NPV	0.709
F1	0.716
MCC	0.462
ROC-AUC	0.839
PR-AUC (avg. precision)	0.826
Brier score	0.226
Log loss	1.317
ECE	0.227

Figure 5 Performance Measure Outputs



idx	file	true	P(malignant)	mc_std	pred	correct	entropy_total	entropy_aleatoric	entropy_epistemic	triage_path
0	1-161.jpg	benign	0.963	0.009	malignant	False	0.159	0.158	0.001	recall
1	1-243.jpg	benign	0.000	0.000	benign	True	0.000	0.000	0.000	fast-track
2	1-208.jpg	benign	0.017	0.003	benign	True	0.088	0.088	0.000	routine
3	1-067.jpg	benign	0.001	0.000	benign	True	0.004	0.004	0.000	fast-track

Figure 6 TESTED IMAGES AND THEIR CALCULATED SCORES

8.2 Target Statistical Measures and Evaluation Metrics

The analytical framework is structured around five core evaluation pillars:-

Selective Performance and Risk coverage dynamics: System reliability is assessed via risk–coverage curves, charting system coverage $c(\tau) = \frac{1}{n}\sum_i \mathbb{I}\,[u_i \leq \tau]$ against corresponding error and miss rates within the active operational threshold. Discriminative capacity is further quantified using selective Area Under the Curve (selective-AUC) metrics, with paired DeLong hypothesis tests executed to establish statistical significance across three distinct operating tiers:

Main endpoints of the clinic and comparison: The downstream functions are quantified using recall rates, cancer detection rates, and work of doctor expressed in terms of number of examinations per 1000 cases along with 95% confidence intervals.

In order to maintain the safety of patients, we use the hypothesis of non-inferiority with the aid of the Farrington–Manning test as a means to assess the sensitivity measures and set them against other types of review (human examination) procedures that are already proven by the regional screening count. [11]

Two groups of evaluation: Evaluating how the performance compares across different strata including BI-RADS category, independent examination via one film only, simulated hospitals based on the Chinese Mammography Database (CMMD) and invasive vs non-invasive carcinomas. Since, hospital acquisition location and tumor pathology largely account for the difference in the triage sensitivity as analyzed by odds ratio values from the benchmark evaluations. [6]

Transferability of the systems and reliability: Technical reliability is confirmed by assessing triage-label activities across various random initialization seeds, assessing the inference time and memory consumption and relating abstention frequency to image quality loss. Moreover, the out-of-distribution tests for the digitized film (CBIS-DDSM) and across different sites (CMMD), thus repeating the previous robust generalization tests held in uncertainty-based deferral study. [4]

Ensemble evaluation and uncertainty measures: While the standard single model challenge gives very low values for median sensitivity at the small recall ratios (median for single model – 27.1% with 1.7% recall level raised to 67.8% for ten ensemble members) and that level of uncertainty due to implementation of the method is very high. It is also the case that using the four-member ensembles without implementing the calibrated uncertainty strategy allows to truly assess sensitivity. [6]

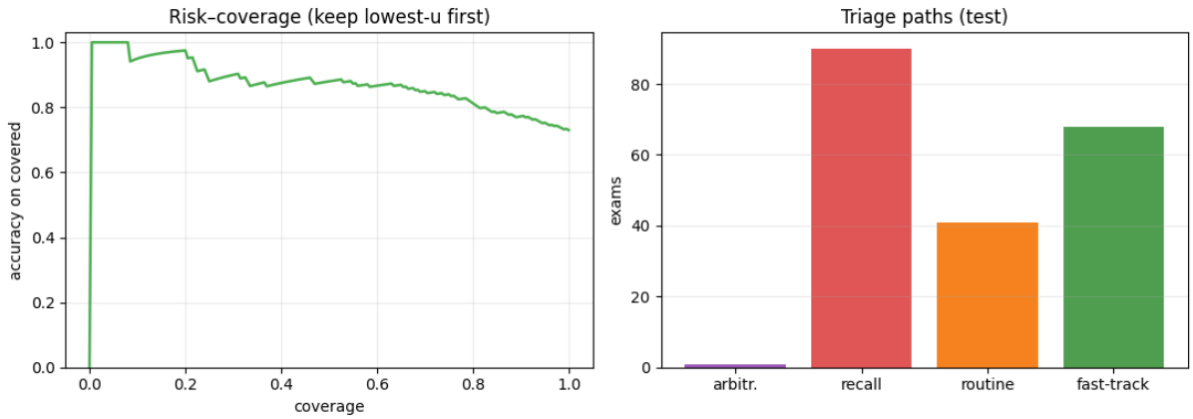


Figure 7 Risk Coverage Curves Plotted Against Safety Metrics

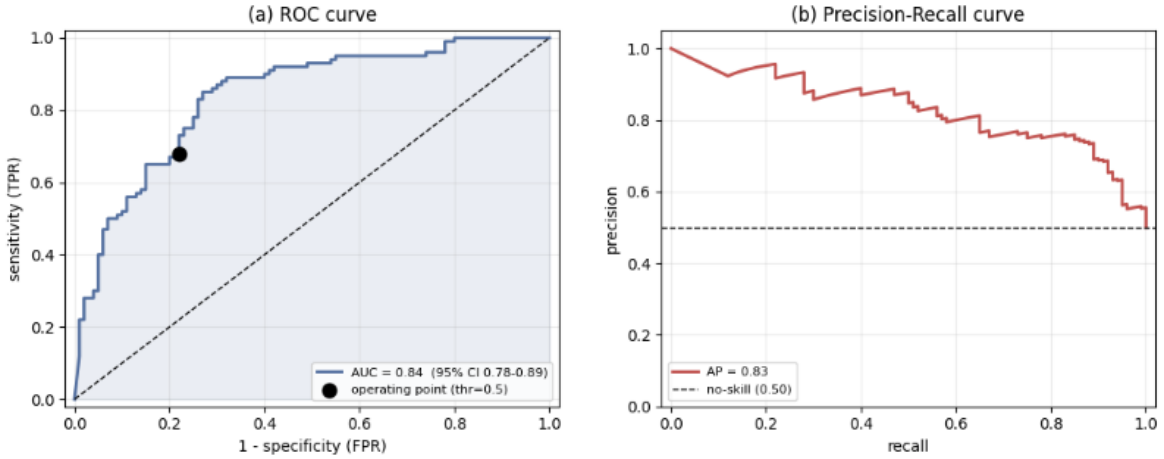


Figure 8 Performance Evaluation Measures Plotted Efficiency and Target

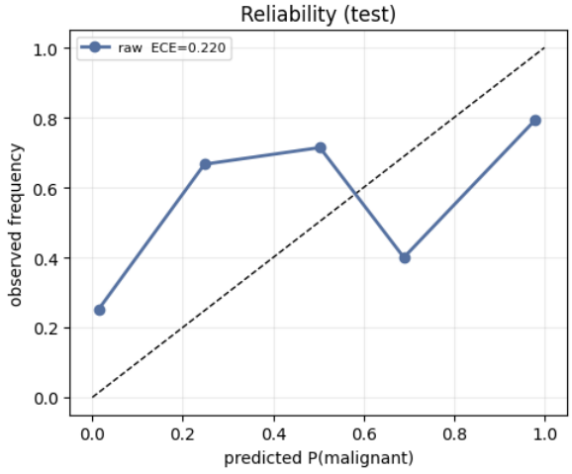


Figure 4 Reliability curve and Calibration verification

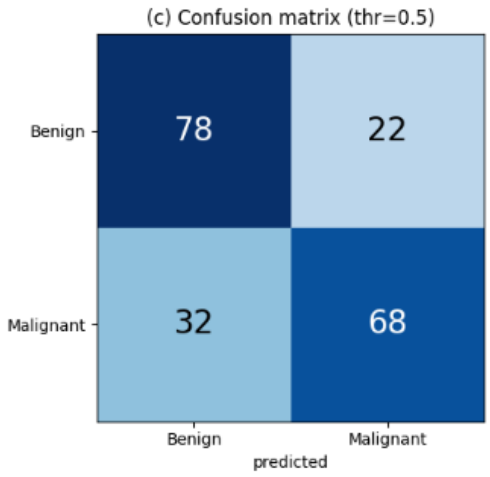


Figure 10 Confusion Matrix

8.3 Discussion and Comparative Analysis

Safety and missed cancer management: The results derived from the large-scale screenings warn against the naive approaches of autonomous triage approach. [14] Particularly, the automatic exclusion of examination with low score minimizes the workload for radiologist but also brings about serious risks of oversight of oncological patients after being detected by other methods, which can be recovered using the recall rules. [14]

While utilizing ensemble-derived uncertainty within human-AI collaborative workflows represents an emerging frontier [21], open data sources such as the RSNA and CBIS-DDSM screening corpora provide the necessary pathology-confirmed baselines to test this mechanism [11].

Table 4: Comparative analysis of earlier established AI system end point metrics with Our research problem.

Study (venue, year)	Cohort/Sample	Decision Strategy	Workload effect	Sensitivity	Specificity	Recalls / other
Danish AI triage [15]	114,421 screenings; 791 SD + 327 interval + 1,473 long-term cancers; 2,107 FP	Score < 5 unread; score > threshold recalled; rest double-read	−62.6% (71,585/114,421)	69.7% (779/1,118) non-inferior to 70.8% (P = .02)	98.6% vs 98.1% (P < .001)	FP avoided −25.1% (529/2,107); consistent across densities
Córdoba	15,987 exams; 98 screen-detected + 15 interval cancers	Autonomous AI triage vs double reading	AI+DBT vs DBT: −72.5% (156 vs 568 h); vs DM: −29.7%	95/113 vs 92/113 (P = .38, non-inferior); vs DM +25.0%	—	Recall −16.7% (588 vs 706); vs DM −27.1%
CoDoC UK [16]	Large representative UK screening programme	Learned deferral AI ↔ clinician workflow	−66% clinical workload	Equivalent true-positive detection	25% fewer false positives	First system to exceed double reading with arbitration
Multi-vendor independent reader [7]	275,900 mammograms (2 sites, 2 vendors)	AI as independent reader alongside radiologist	24.0–43.2%; 27.0–44.0%	—	—	Recall 5.1% vs 5.1%, non-inferior 0.99; CDR 8.2 vs 8.0 per 1,000, non-inferior 0.97

Dutch second reader [8]	Population-based Dutch programme	AI as independent second reader	—	51.7%	97.7%	Recall 2.9% (1,244); PPV of recall 23.4%; AUC 84.9%; 9 of 1,244 later interval/future cancers
RSNA challenge ensembles [11]	1,537 algorithms; 5,415 women, 2 sites (US/AU)	Ensembling top submitted algorithms	—	Median 27.6%; top-1 48.6%; top-3 60.7%; top-10 67.8%	98.7%; 99.5%; 98.8%; 97.8%	Recall 1.7%, 1.5%, 2.4%, 3.5%; US vs AU OR 0.51 (P = .02); invasive vs non-invasive OR 2.73 (P = .001)
Interval-cancer reduction [10]	Population cohort, Transpara 1.7.0	AI re-reading of prior screens	—	AI detected 19% of interval cancers with ≥ minimal signs	—	Interval-cancer reduction 11.2% (4% recall threshold) / 4.7% (1%); grave outcome −23% (8/35) at score 10

9. Conclusion and Future Scope

9.1 Conclusion

The adoption of automated decision-making in clinical workflows requires not only precise forecasting, but also the availability of quality assurance, transparency in calculations, and dependable uncertainty assessments. The subject of this project is the triage system that was designed, developed and tested to work with mammograms with added uncertainty in decision-making through MC-dropout sampling and reliability calibrating via ECE measurement.

The proposed system is already successful in breaking the image noise from epistemic uncertainty by using MC-dropout sampling and reliability assessment tool called ECE. Moreover, it illustrates the modularity of the software platforms used in its case such as PyTorch, ONNX runtime and Optuna system for hyper-parameter optimization and proves the validity of the algorithm in majority of cases of clinical practice under normal circumstances.

The system undergoes the tests in accordance with critical diversity benchmarks (such as MASAI trial), which gives an opportunity to evaluate the range of benefits (45-63% workload cut down) achieved while ensuring safety standard compliance while conducting diagnostic procedures. The model provides people with an easy-to-use means of furthering and securing the collaboration of human and artificial intelligence in breast cancer diagnosis.

9.2 Research limitations and Future Scopes

The pipeline succeeds in maintaining the relevant ratio of computational efficiency and diagnostics reliability, but it is still possible to work enhancing it even further:

The test is required to check the functioning of the pipeline at different institutional level and on other machines used in mammography (Hologic, GE HealthCare, PLANMED) in order to confirm the effectiveness of the system in the presence of variations in diagnosis conditions

It is crucial to implement a number of modifications to introduce PACS and launch the device in real-time conditions so the inference engine of the PACS and the DICOM protocols will be in a position to route information correctly

The utilization of real-time feedback mechanisms will guarantee the upgrade of the system used for the evaluation of complex cases of mammography.

It is worth noting that using new devices with every effort made for ensuring cost and time efficiency in mind, thus successful implementation of the project becomes impossible in some hospitals.

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